flyPad Material 3 redesign and independent flyPads

A new Material 3 look for every flyPad page, a flyPad per pilot on both aircraft, a custom flyPad theme, the A32NX buffet onset (coffin corner) page, GSX services link fixes, flyPad fixes, a landing calculator that keeps calculating above the wind limits, the RAT stow and cargo door rows of the Services page, pushback and A380X door button fixes, the A32NX MCDU engine-out REC MAX, the A380 taxiway width check and the A380X smart FIVE HUNDRED pin option

Aircraft	A32NX and A380X (the Buffet tab: A32NX only)
Where to find it	The captain and first officer flyPads (all pages)
Branch	`core/efb-redesign`, based on `core/efb-ground` (stacked: needs `core/surveillance`, `core/a380x-mfd-fms`, `core/efb-performance` and `core/efb-ground` merged first). The independent flyPads need two companion commits in the `aircraft-large-files` repository: 24f7b02 (A380X) and d3060ea (A32NX). The RAT stow row needs `core/rat-ground-stow` (the systems side of the stow); the A380X cargo door row reads the inhibition written by `core/a380x-cargo-doors` (merge both). The smart FIVE HUNDRED option needs the callout of `core/a380x-elec-fuel-alerts` (merge both).
Status	Sim-tested by the user: Services and Pushback look, quick controls fix, A32NX independent flyPads, frame and Dashboard / Dispatch / Performance / other pages on both aircraft. Fuel / Payload / Taxi look approved by the user. Not sim-tested (browser harness only): the flyPad UI fixes, the Payload LOAD row change, the checklist and lighting preset fixes, the landing wind check. RAT stow row: sim-tested on both aircraft. Tested in the sim by the user (a combined test build, not this core branch itself): Buffet tab on the A32NX and its Live / What if pill row fix (A32NX), custom theme (A32NX, user word), the first colour wheel ("the wheel works"), GSX boarding passenger count (A32NX), GSX turnaround Done memory, Baggage row Done, Turnaround scrollbar and tug slider fixes (A380X). A380X cargo door row: sim-tested with the systems part (PASS, recorder-checked). Pushback map speed-up and pushback going on off the page: confirmed in the sim by the user. A380X Main 5 R / Main 4 R door buttons: sim tests passed. A32NX MCDU engine-out REC MAX: tested in the sim by the user: OK. The vivid colour wheel that replaced the first one: OK in the sim (A380X). GSX refuel progress and jet bridge state: OK in the sim (A380X, GSX 4). Not sim-tested: the Light theme label fix, the greyed GSX switch, the GSX deboarding passenger number, and the A380 taxiway width check and smart FIVE HUNDRED option (added).
Pictures	No pictures: the flyPad pictures of the older manuals show the look before this redesign.

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1. Material 3 look of the Ground pages

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground | **Status:** Services and Pushback tested in the sim by the user: OK. Fuel, Payload and Taxi: approved by the user, checked in the browser harness. Pushback map drag fix: not sim-tested on its own; the

> later map speed-up (only the visible airport, at most 10 Hz) and the pushback going on off the page: confirmed in the sim by the user ("pushback fixed"). Tug speed slider fix: confirmed in the sim by the user (A380X). A380X Main 5 R / Main 4 R door buttons: sim tests N1 and P1 passed. Unit tests: pushbackAirportDraw.spec, pushbackControllerInput.spec, A380ServiceDoors.spec, groundServicePage.spec.

What it is for

The flyPad pages get a Material 3 dark look on the flyPad cyan: tonal cards, dense list rows with switches and chips, a segmented control for the tabs. A shared Material kit (page, cards, section headers, list rows, status and action chips, switch, segmented control, buttons, progress bar, banner) on new theme tokens for the blue, dark and light themes is used by every redesigned page. The Ground pages came first. The functions of the pages do not change.

What you see

Page	New look
Services	Three columns: the doors and the ground equipment as scrolling lists (switch or Request / Release chip, GSX status and fuel progress on the rows); the aircraft from above, nose up and zoomed on the fuselage, its doors coloured with their state, the holds and service tags (A380X two decks, A32NX one); the GSX card (switch, turnaround timeline of the departure or arrival with its Baggage step, other GSX services as chips, GSX questions and messages; see "GSX services link" and "GSX turnaround" below). Status chips: GPU, tug, doors open, chocks and cones (A32NX).
Pushback	The map fills the left card. A control rail: system switch, Call / Release tug button, parking brake row, Backward / Paused / Forward, a steering dial over the direction slider, the speed slider, Left / Straight / Right and the controller switch. When the aircraft is at the SimBrief origin or destination, the Navigraph airport map (aprons, taxiways, runways, guidance lines, stands) is drawn under the pushback map, heading up.
Fuel	The aircraft with its tanks on a wide card and a refuel rail (target, slider, start and stop, duration, plan summary). A380X: the whole aircraft outline with the tank quantities in tables (each feed tank now shows its own quantity; before, the four feed tanks of the drawing read feed tank 3). A32NX: the over-wing outline with its tanks filled, the quantities as tiles.
Payload	The cabin and seat map on a full-width card with the holds under it (load over capacity, click to plan), then the load table, the boarding controls and the balance chart.
Taxi	The route form as cards, the runway crossings as banners, the map buttons and the frequency box in the kit.

Pushback map fix: the airport under the pushback map redrew only after the map centre had moved about 100 m, so the aircraft slid over a frozen airport while the map was dragged. It now follows every move.

Pushback map: only the visible airport, at most 10 times a second

The airport layer drew the whole Navigraph airport (thousands of points) on every position update, 20 times a second: the A380X ran at 21 FPS on the Pushback page against 42 FPS on another page (capture). It now draws only the shapes and lines that reach the map window, and redraws only when the picture moves by about a pixel, at most 10 times a second.

The pushback goes on when another flyPad page is opened

Leaving the Pushback page stopped the tug (with a short "Pausing Pushback" message), because the rudder pedal and sidestick steering lived in the page. The steering now runs in the captain flyPad itself, whatever page is shown (when the controller switch of the Pushback rail is on). The page resets the tug commands only when it opens with no tug attached (captain flyPad), so a pushback in progress goes on when the page is left and opened again.

A380X Services door buttons: Main 5 R and a new Main 4 R

The aft right door button said "Main 4 R" but opened MAIN 5R, the door the SD DOOR page and the FWS name. It is now labelled **Main 5 R** and its marker on the planform sits on door 5 R. A new **Main 4 R** button opens and closes MAIN 4R. The Doors card lists Main 1 L, Main 2 L, Upper 1 L, Main 4 R, Main 5 R and the cargo row: six rows without a

scrollbar. The catering truck still docks at MAIN 5R: a button only moves its door; the door a ground vehicle (or GSX) docks at comes from the aircraft configuration.

Tug speed slider fix: the handle of the vertical tug speed slider of the Pushback rail sat 5 px right of its rail (the slider library offset suits its own 14 px handle, the flyPad handle is 24 px). It is now centred on the rail.

Limitations and why

Limitation	Why	What to do
The airport under the pushback map needs the Navigraph airport map.	It is drawn from the Navigraph airport database, only at the SimBrief origin or destination.	Link a Navigraph account and import a SimBrief flight plan.
Spacing is made with margins, not the CSS gap.	The sim browser (Coherent GT) ignores the CSS gap property.	-

How to use

1. Open **flyPad > Ground** and pick the tab (Services, Fuel, Payload, Pushback, Taxi) at the top right.
2. Use the switches, chips and buttons as before: open or close a door, request a service, refuel, board, push back.

How to test

#	Do this	Expected result
1	A380X at a gate, aircraft powered, SimBrief flight plan imported. Open flyPad > Ground > Services.	Three columns: DOORS and ground equipment lists, the aircraft from above (two decks) with its doors, the GSX card. Nothing cut at the edges.
2	Open a door with its switch in the list.	The row status changes and the door shape on the aircraft drawing changes colour.
3	Repeat on the A32NX.	Same layout with the single-deck A320 drawing and the chocks and cones chips.
4	Open Ground > Pushback (Navigraph linked, aircraft at the SimBrief origin).	The map on the left with the airport under it; the control rail on the right.
5	Drag the pushback map.	The airport moves with the aircraft symbol at once (no sliding over a frozen airport).
6	A380X: compare the frame rate (MSFS FPS display) on the Pushback page and on the Dashboard.	About the same frame rate on both pages (before the fix, about half on the Pushback page).
7	Call the tug and push back with the controller switch on (sidestick back, rudder pedals to steer). During the push open the Dashboard and steer with the rudder pedals.	The pushback goes on and follows the rudder pedals; no "Pausing Pushback" message. Open Pushback again: the push goes on.
8	A380X at a stand, engines off: Ground > Services, Doors card. Switch Main 5 R on, then ECAM control panel DOOR key.	The SD DOOR page shows MAIN 5R open (the same door); the 5 R marker on the planform lights up. Switch it off.
9	Switch Main 4 R on, wait until open, then off.	The 4th right main deck door opens and closes; Main 5 R stays closed; the Doors card shows six rows, no scrollbar.
10	Look at the vertical tug speed slider on the Pushback rail and move it.	The handle is centred on its rail over the whole travel.

11	Open Ground > Fuel on the A380X, then on the A32NX.	A380X: whole aircraft with tank tables, each feed tank with its own quantity. A32NX: wing tanks filled, quantities as tiles. The refuel rail on the right.
12	Open Ground > Payload, then Ground > Taxi.	Payload: cabin card across the page, holds under it, load table, boarding and balance cards. Taxi: route cards and map in the new look.

2. Material 3 look of the frame and of the other pages

> **Aircraft:** A32NX and A380X | **Where:** flyPad frame; Dashboard, Dispatch, Performance, Navigation, ATC, Failures, Checklists, Presets, Settings | **Status:** Tested in the sim by the user on both aircraft: OK, incl. the quick controls on both tablets.

What it is for

The rest of the flyPad in the same Material look, so that every page matches. The shared switch, segmented selects, text fields, drop-downs and tooltips take the new look, so every page follows.

What you see

Part	New look
Frame	A navigation rail with the icon names (the open section in a tonal pill), the status bar, the quick controls as a card (brightness sliders with their Auto switch, state tiles, sim rate tile at the bottom right), the section tabs as a segmented control.
Dashboard	The flight card (airports, progress, figure tiles, route, SimBrief import) and the Important Information column (weather readings or raw report, pinned charts, maintenance and checklists as rows).
Dispatch	Overview: the nose drawing on a wide card and the aircraft figures in three lists; OFP and Printouts in kit cards and buttons.
Performance	A shared calculator kit (form cards, labelled rows, result tiles, the fill-from button): takeoff, landing and descent results as tiles; temperature correction with one row per published altitude and its corrected value beside it. A32NX: a new Buffet tab (see "Performance > Buffet" below).
Navigation	Chart list, viewer controls and pinned charts in the kit.
ATC	Type filter chips, controller cards with their buttons always shown, the COM 1 card.
Failures	Chapter cards with failure and active counts, failure tiles.
Checklists	The list with done / next marks and counts, the open checklist with its progress.
Presets	State tiles with the loading card; lighting presets beside the automatic loading.
Settings	Two panes: the categories beside the open page. Settings > flyPad has the theme choice, with a custom theme (see "Custom flyPad theme" below).

How to use

1. Pick a section on the navigation rail on the left, then a tab in the segmented control at the top right.
2. Open the quick controls from the status bar (brightness, Auto, state tiles, sim rate).

How to test

#	Do this	Expected result
1	Either aircraft, flyPad on. Look at the left rail.	Icons with their names; the open section in a cyan pill.

2	Open the quick controls from the status bar.	A card with the brightness slider and its Auto switch, state tiles and the sim rate tile at the bottom right.
3	Open Dashboard, then Dispatch > Overview, OFP, Printouts.	Cards in the new look; the A380X / A320 nose drawing on the Overview.
4	Open Performance > Takeoff, Descent, Landing, Temperature Correction.	Form cards and result tiles; the temperature correction lists one row per altitude with its corrected value.
5	Open Navigation, ATC, Failures, Checklists, Presets.	Every page in the new look, no old grey boxes.
6	Open Settings and pick a category.	Two panes: categories on the left, the open page on the right.
7	Repeat on the first officer flyPad (see "Independent captain and first officer flyPads").	Same look; the quick controls work there too.

3. flyPad UI fixes after the redesign

> **Aircraft:** A32NX and A380X | **Where:** Settings, dialogs, toasts, Ground pages, texts | **Status:** Checked in the browser harness on both aircraft. Not sim-tested. Unit tests: detent gauge, Settings categories, fuel percentage, failure counts, descent notes.

What it is for

A check of every flyPad page on both aircraft and both tablets after the redesign found 23 problems: three layout bugs and parts still in the old look or with poor texts. They are fixed:

Area	Fix
Throttle calibration	Opened inside the two-pane Settings it was taller and wider than the pane: the button bar was cut and, with 4 axes on the A380X, Axis 4 and Save and Apply were off screen. It is now one card that fits, the axes in one row, the button bar always visible, a new detent gauge. The in-detent highlight also works at exactly 0 (its colour is now the light primary, not green).
Settings	About: the content covered the title, now in the page flow with build info rows. The open Settings category gets its icon colour and bold label (the active class was lost). Navigraph sign-in, SimBrief trash, Aircraft Options Select, Automatic Call Outs Reset and the Sim Options Calibrate button and SimBridge status in the kit look; language and keyboard selects sized like the others.
Dialogs and shared parts	Confirmation and alert dialogs, scroll buttons, progress bar, card, divider, the critical error page and the toasts (each toast type its own colour, errors red) no longer use the old colours.
Ground pages	Services legend at its small size; Taxi frequencies button and panel, empty-state text, one-line database chip, Entry chevron; Payload balance chart in the new colours and font; Fuel: no "NaN %" or full bars while the tank capacities are still 0.
Texts	Presets help in cards, with the A380X wording (MFD with KCCU, AFS CP); Performance tab "Descent"; "1 active failure", "1 failure"; descent notes without a trailing separator; the A380X descent legend names the A320 FCOM anti-ice source; chart list and pinned charts empty texts; ATC network name in capitals and an empty controller list message.

Limitations and why

Limitation	Why	What to do
A tooltip ("Click Text to Change the Preset's Name") can stay on screen after leaving the Presets page under the mouse.	An older flyPad bug, not part of these fixes.	Move the mouse over the flyPad again.

How to use

1. Nothing to set. Calibrate the throttle from **flyPad > Settings > Sim Options > Calibrate** as before.

How to test

#	Do this	Expected result
1	A380X: flyPad > Settings > Sim Options > Calibrate.	The calibration fits the pane: four axes in one row, the button bar with Save and Apply visible.
2	Same on the A32NX.	Two axes, the button bar visible.
3	Move a throttle into a detent, incl. the 0 (idle) detent.	The detent is highlighted, also at exactly 0.
4	Settings > About.	The title is not covered; build info as rows.
5	Click several Settings categories.	The open category has a coloured icon and a bold label.
6	Trigger a confirmation dialog (e.g. Ground > Payload deboarding) and a toast.	Both in the Material colours; an error toast is red.
7	Ground > Fuel right after loading the aircraft.	No "NaN %" and no full bars before the tank capacities are known.
8	Failures page with one failure active.	The text reads "1 active failure" (singular).
9	Performance tabs.	The descent calculator tab is named "Descent".

4. Payload page: LOAD title row and balance chart

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Payload | **Status:** Checked in the browser harness on both aircraft; the user reviewed the first version in the sim (screenshot) and chose the grid end. Final version not sim-tested. Unit tests: chart fit.

What it is for

On the redesigned Payload page the LOAD title sat on its own line above the load table and the balance chart left an empty band in its card. Now:

- The LOAD title sits on the PLANNED / CURRENT header row of the load table.
- The SimBrief fill action is a full-width button under the table, as on the Fuel page. It shows with the same rule as before (an OFP loaded that differs from the planned load, no boarding or GSX payload sync running).
- The balance chart fills the height of its card (never smaller than the former 280 px).
- The weight grid ends at the lowest labelled weight (A32NX 40 t, A380X 270 t) instead of leaving an empty band under the envelope; the weight labels are centred on their grid lines, the unit sits under the lowest label, and the MTOW / MLDW / MZFW names are re-placed for the new scale.

Limitations and why

Limitation	Why	What to do
A full MSFS restart is needed after an update.	The chart scale lives in the aircraft config files (flypad-payload.json5).	Restart the sim after installing.

How to use

1. Open **flyPad > Ground > Payload**; plan the load in the table or with the SimBrief button under it.
2. Read the centre of gravity on the balance chart.

How to test

#	Do this	Expected result
1	A32NX at a gate, flyPad > Ground > Payload.	LOAD is on the PLANNED / CURRENT row of the load table.
2	Import a SimBrief flight plan whose load differs from the planned one (no boarding running).	A full-width SimBrief button appears under the table.
3	Look at the balance chart.	It fills its card; the lowest grid line is 40 t, labels centred on their lines, the unit under the lowest label.
4	Repeat on the A380X.	Same layout; the lowest grid line is 270 t.

5. Independent captain and first officer flyPads

> **Aircraft:** A32NX and A380X | **Where:** Captain and first officer tablets | **Status:** A32NX tested in the sim by the user: OK. A380X: no separate record; both tablets were used in the sim test of the redesigned pages (quick controls on both tablets OK). Unit tests: tablet index.

What it is for

Both tablets showed the same flyPad: their screens share one material in the cockpit model and one gauge drew on it, so one pilot changing a page changed it for the other. Now the first officer tablet has its own flyPad gauge (`efb.html?Index=2`), with a new material on its screen in the cockpit model.

What is per tablet and what is shared

Per tablet	Shared
The page shown and the navigation	The flyPad settings
The loaded OFP (SimBrief import)	The aircraft state (doors, services, fuel, payload)
The checklists and their progress	
The brightness and its Auto setting (first officer: `L:A32NX_EFB_2_BRIGHTNESS`, settings `EFB_2_*`)	
A32NX: the power button (first officer: `H:A32NX_EFB_2_POWER`; before, both buttons sent the same event, so one press toggled both tablets)	

The actions that change the shared aircraft state by themselves run on the captain flyPad only: the tug release after a pushback, the automatic loading of lighting presets, GSX remote control, the rudder pedal and sidestick steering of the pushback, and the reset of the tug commands when the Pushback page opens with no tug attached. A flyPad clears only the OANS taxi flags it has set itself.

Limitations and why

Limitation	Why	What to do
Without the aircraft-large-files companion, the first officer tablet keeps showing the captain flyPad.	The new screen material is in the high-detail cockpit model, which lives in aircraft-large-files (A380X 24f7b02, A32NX d3060ea); the large-files pointer of this branch is not bumped yet.	Merge the two large-files commits, then bump the `large-files` pointer.
A full MSFS restart is needed after an update.	panel.cfg, the cockpit behaviour and the model change.	Restart the sim after installing.

Lighting presets load automatically from the captain flyPad only.	Two flyPads would load them twice.	Keep the captain flyPad on.

How to use

1. Use each tablet on its own: open any page, import the OFP, run the checklists.
2. Set each tablet brightness from its own quick controls.
3. A32NX: each tablet has its own power button.

How to test

#	Do this	Expected result
1	A380X with this branch and the large-files companion, MSFS restarted. Turn both tablets on. Open Dashboard on the captain tablet and Ground > Fuel on the first officer tablet.	Each tablet shows its own page.
2	Import the SimBrief OFP on the captain tablet only.	The captain Dispatch > OFP shows it; the first officer OFP stays empty until imported there.
3	Change the brightness in the quick controls of the first officer tablet.	Only that tablet changes.
4	Open a checklist and check items on one tablet.	The other tablet checklist is unchanged.
5	Repeat on the A32NX; then press the power button of the right tablet.	Only the right tablet turns off.
6	Push back with the flyPad, then open and leave the Pushback page on the first officer tablet during the push.	The pushback does not stop (leaving the page stops it on neither tablet).

6. Quick controls clicks no longer open the troubleshooting page

> **Aircraft:** A32NX and A380X | **Where:** Status bar > quick controls | **Status:** Tested in the sim by the user: OK.

What it is for

The quick controls pane is drawn inside the status bar. When the status bar shows an issue (red), a click on it opens the troubleshooting page, so every click in the quick controls, and the click that closes them, also opened that page. The pane and its backdrop now keep their clicks, and the backdrop closes the pane on the click itself (closing on mouse down let the click land on the status bar under it).

How to use

1. Open the quick controls from the status bar, use them, close them with a click outside.
2. To open the troubleshooting page, click the red status bar itself.

How to test

#	Do this	Expected result
1	A flyPad with a red status bar (an issue listed in Settings > About > Troubleshooting). Open the quick controls.	The quick controls open.
2	Click a switch or slider in the quick controls.	The control works; the page under it stays (no troubleshooting page).
3	Click outside the quick controls.	They close; the troubleshooting page does not open.

4	Click the red status bar.	The troubleshooting page opens, as before.

7. Checklists: always loaded, green check when marked complete

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Checklists | **Status:** Checked in the browser harness. Not sim-tested. Unit tests: checklist marks.

What it is for

- **Checklists sometimes missing.** The checklist file was read without waiting for its text, so an empty list was returned; the flyPad stored (and froze) it and the checklists parsed later could not be added. The file is now read and parsed before the list is returned; a second call waits for the same read, and a failed read is tried again by the next call.
- **Green check.** The list showed the done mark only for the checklists of the current or previous flight phases, so with the aircraft unpowered (flight phase 0) a completed checklist kept an empty circle. A checklist marked with **Mark checklist as complete** now shows a filled green check, whatever the flight phase. One with all items done but not marked keeps the empty circle.

How to use

1. Open **flyPad > Checklists**, pick a checklist, check its items.
2. Press **Mark checklist as complete** at the end.

How to test

#	Do this	Expected result
1	Load the aircraft several times (new flights) and open flyPad > Checklists each time.	The checklists are always listed.
2	Aircraft cold and dark (unpowered). Open a checklist, check all its items, but do not mark it.	The list shows an empty circle for it.
3	Press Mark checklist as complete.	The list shows a filled green check for it, although the aircraft is unpowered.

8. Automatic lighting preset loading lists all eight presets

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Presets > Interior Lighting | **Status:** Checked in the browser harness. Not sim-tested. Unit tests: preset list.

What it is for

The Day, Dawn/Dusk and Night drop-downs of the automatic preset loading offered only the presets that had a name: with unnamed presets they offered only "None". They now list the eight presets as "<number> - <name>" ("No Name" when unnamed).

How to use

1. Open **flyPad > Presets > Interior Lighting** and save the cockpit lighting into a preset.
2. Turn on the automatic loading and pick a preset for Day, Dawn/Dusk and Night.

How to test

#	Do this	Expected result

1	flyPad > Presets > Interior Lighting with no preset named. Turn on the automatic loading and open the Day drop-down.	None, then 1 - No Name ... 8 - No Name.
2	Name preset 2 "Taxi" and open the drop-down again.	It lists 2 - Taxi among the eight.
3	Pick 2 for Day, reload the flight in daytime with the aircraft powered.	Preset 2 is loaded by the captain flyPad.

9. Error details on the flyPad error screen

> **Aircraft:** A32NX and A380X | **Where:** flyPad error screen | **Status:** Used in the sim by the user: it showed the error of the Top of Descent crash, which was then fixed.

What it is for

The flyPad error screen only said that something went wrong. It now shows the error (name and message), the first lines of its stack and the React components it was thrown in, so a crash can be reported and found.

How to use

1. If the flyPad shows the error screen, take a screenshot of the error box and report it.
2. Press **Reset Display** on the error screen to recover.

How to test

#	Do this	Expected result
1	There is no normal way to crash the flyPad on purpose. When the error screen appears (or in the browser harness with a forced error), look at it.	Under "A critical error has been encountered.": a box with the error name and message, up to 6 lines of its stack and up to 6 lines of the component stack.

10. Landing calculator above the wind limits, with a wind check

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Performance > Landing | **Status:** Checked in the browser harness on both aircraft. Not sim-tested. Unit tests: wind ranking, calculators beyond the tailwind limit.

What it is for

A crosswind above the maximum asked "Calculate anyway?" and a tailwind above the limit gave an error and no result. Now the calculation always runs: the pilot sees the figures with a warning and can compare the other runways and the alternate. This is on this branch because it changes the redesigned performance widgets.

What you should see

Situation	What you should see
Crosswind above the maximum	Results shown, with a red banner "Crosswind *n* kt, maximum *m* kt": above the maximum for this runway condition, figures for information.
Tailwind above the limit	Red banner "Tailwind *n* kt, limit *m* kt". The distances, stop margin and MLW(PERF) continue the tailwind correction of the data and are marked extrapolated (~ and a dagger, amber), with a note under the results.
Wind check tab (beside Landing roll, red dot when the wind is above a limit)	The destination runways ranked by wind: within limits first, most headwind on top; LDA, HEAD/TAIL, CROSS and LIMITS columns; each with a **Use** button that loads it.

Alternate card in the wind check	The OFP alternate (ICAO editable), its METAR from the weather source of the settings and its best runway, with **Calculate for <ICAO> <runway>**.
Wind entered without a direction	The ranking asks for the wind with its direction (e.g. 330/45).

The limits used

	A32NX	A380X
Maximum crosswind, gust included	Dry, good or wet 38 kt; 29, 25, 20 or 15 kt on contaminated runways (A320 FCOM PER-LDG-DIS-MAT)	Dry / wet 40 kt; 20 kt slush or dry snow; 15 kt standing water or compacted snow; 5 kt icy
Tailwind limit	15 kt	10 kt

Gusts are included in the components as in FCOM LIM; a variable wind is taken from the worst direction.

Limitations and why

Limitation	Why	What to do
The wind check is not an Airbus function.	It is a design choice of the flyPad: advice only, the flight crew decides.	-
Figures beyond the tailwind limit are extrapolated.	The Airbus data stop at the limit; the tailwind correction is continued.	Use them as an indication only.
The alternate check needs an internet connection and the OFP.	The METAR comes from the weather source set in the flyPad settings; the alternate from the SimBrief OFP.	Enter the alternate ICAO by hand if the OFP has none.

How to use

1. Fill the landing calculator as usual and press Calculate.
2. If a red wind banner shows, open the **Wind check** tab beside the landing roll picture.
3. Load a better runway with **Use**, or the alternate with **Calculate for**, and calculate again.

How to test

#	Do this	Expected result
1	A380X, SimBrief OFP with an alternate loaded. flyPad > Performance > Landing. Enter a runway, a dry runway condition and a wind giving about 45 kt of crosswind. Calculate.	Results shown; red banner "Crosswind 45 kt, maximum 40 kt"; red dot on the Wind check tab. No Yes/No prompt.
2	Enter a wind giving a 15 kt tailwind. Calculate.	Results shown with "Tailwind 15 kt, limit 10 kt"; distances marked ~ and a dagger in amber, and the extrapolation note.
3	Open the Wind check tab.	The destination runways ranked, those within limits first; LIMITS column says "Within limits" or which limit is exceeded.
4	Press Use on the best runway.	The runway is loaded into the calculator; calculate again: no banner if it is within limits.
5	Look at the alternate card.	The OFP alternate, "Fetching the METAR..." then its METAR and best runway; "Calculate for <ICAO> <runway>" loads it.

6	Repeat on the A32NX with a 16 kt tailwind.	Banner "Tailwind 16 kt, limit 15 kt", extrapolated figures.

11. RAT stow (maintenance) row on the Services page

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Services, equipment list | **Status:** Tested in the sim by the user: OK on both aircraft. Unit test: ratStow.spec (row look and status text).

What it is for

The flyPad side of the RAT ground stow: a maintenance row that asks the systems to stow the RAT after the landing (A320 FCOM DSC-29-10-20: the RAT "can be stowed only when the aircraft is on the ground"). The systems side (`L:A32NX_RAT_STOW_REQUEST`, the shared Rust RAT model) is on `core/rat-ground-stow`; see that manual for the rules, the limitations and the full test.

> **CAUTION**
> Merge both branches: without `core/rat-ground-stow` the row shows while the RAT is out but pressing Stow has no effect.

What you should see

Situation	What you should see
RAT stowed	No RAT row.
RAT out, aircraft stopped on the ground	"RAT stow (maintenance)", status "RAT extended", button **Stow**.
RAT out, in flight or moving	Row greyed, status "RAT extended · stow when stopped on the ground".
Stow pressed	Status "Stowing" for about 3 s; the row disappears when the systems have stowed the RAT, or stays if they refused (an extension still commanded).

How to use

1. Stop on the ground with the RAT out and normal electrical power back.
2. flyPad > Ground > Services: press **Stow** on the row "RAT stow (maintenance)".

How to test

#	Do this	Expected result
1	A380X on the ground, engines running: EMER ELEC PWR panel RAT MAN ON (then release it). flyPad Services.	Row "RAT stow (maintenance)", active.
2	Press Stow.	"Stowing", the RAT rig retracts, the row disappears.
3	A32NX on the ground: HYD panel RAT MAN ON (or EMER ELEC PWR MAN ON), then Stow on the flyPad.	The HYD page RAT goes back to stowed, HYD RAT FAULT clears, the row disappears.

12. A380X cargo door row shows the real doors

> **Aircraft:** A380X | **Where:** flyPad > Ground > Services, Doors list (row "Cargo fwd + aft") and the aircraft planform | **Status:** Sim-tested by the user with the systems part of `core/a380x-cargo-doors` on a combined test build: PASS, recorder-checked. Open from 98 %: sim-tested OK by the user with GSX cargo loading (2026-10-17). Unit test: cargoDoor.spec (12 cases).

What it is for

The cargo door row of the Services page read the open request of the ground crew, not the doors. With an engine running the request could be on while the doors did not move, and the row said "open". The row now shows the real

forward and aft doors (position and lock, from the systems) and says when the request waits for hydraulic power.

> **CAUTION**
> Merge both branches: the row reads `L:A32NX_HYD_CARGO_DOOR_OPERATION_INHIBITED`, which only `core/a380x-cargo-doors` writes (without it the row reads 0 = not inhibited and shows the real doors only). That manual describes the systems rules (A380 FCOM DSC-29-10, DSC-52-30-40-10) and the full test.

What you should see

Situation	What you should see
Both doors closed and locked, hydraulic power available	Row "Cargo fwd + aft", status "Closed", switch off.
Engine running, doors closed	Status "Closed · no hydraulic power, engine running"; the switch is greyed out.
Engine running, an open request is on and the doors are closed	Status "Closed · opens once the engines are stopped". The switch can still cancel the request.
Doors moving	Status "Moving" with a progress bar; the FWD and AFT holds on the planform follow the position of each door.
Doors open (from 98 %, where the systems lock them open)	Status "Open". Below 98 % the status is "Moving" (before: "Open" from 90 %).
A door stopped open or unlocked by an engine start	Amber row, status "Not locked · no hydraulic power, engine running"; the switch can still ask the door to close (it follows once the engines are stopped).

The row stays one row for both doors (the request moves both); it is closed only when both doors are locked and open only when both are open. The planform now shows the FWD and the AFT cargo door as two holds, each with its own position.

How to use

1. Stop the engines.
2. flyPad > Ground > Services > Doors: switch "Cargo fwd + aft" on to open, off to close; read the status line.

How to test

Use the same sim test as `core/a380x-cargo-doors`, with both branches installed.

#	Do this	Expected result
1	A380X on the ground, APU on, engine 1 running at idle. flyPad > Ground > Services > Doors.	"Cargo fwd + aft": "Closed · no hydraulic power, engine running", switch greyed out.
2	ENG MASTER 1 OFF, wait for the stop. Switch "Cargo fwd + aft" on; wait 50 s.	"Moving" with a progress bar, then "Open"; the FWD and AFT holds light up on the planform.
3	Doors open: start engine 1 and switch the row off.	The doors hold; amber "Not locked · no hydraulic power, engine running"; the switch stays usable.
4	ENG MASTER 1 OFF and wait until the engine has stopped.	The doors close; the row reads "Closed".

13. Custom flyPad theme, and a readable Light theme

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Settings > flyPad > Theme | **Status:** Custom theme: tested in the sim by the user: OK by the user's word (A32NX, no recording). The first colour wheel (12 hues on 4 rings, alert hues hatched): confirmed in the sim by the user ("the wheel works"). The vivid colour wheel that replaced it: OK in the sim (A380X, "the wheel works"). The Light theme

> label fix: not sim-tested (harness only). Unit tests: themePalette.spec, colourWheel.spec (contrast, alert hue note, stored red / yellow themes kept, presets unchanged).

What it is for

The flyPad had three fixed themes: Blue, Dark and Light. A fourth choice, **Custom**, lets the pilot pick the colours: a base (the surfaces and text of the Gray, Black or Light theme) plus a primary and a secondary colour. The Blue, Dark and Light themes keep their exact colours.

- **Primary colour**: buttons, switches, progress bars, chart lines.
- **Secondary colour**: selected tabs, segmented buttons, chips and the navigation rail. "Same" uses the primary colour.
- The flyPad adjusts only the lightness of the picked colours until every text and control pair reaches the WCAG 2.1 AA contrast (4.5:1 for text, 3:1 for controls). A contrast list beside a preview card shows each pair with "AA" (or "FAIL").
- Amber and red are the alert colours: the preset discs offer none, and a colour close to their hues (picked on the wheel or typed) is accepted with a note that buttons and switches in it may look like warnings.
- The pencil button at the end of the colour discs opens the **colour wheel**: the 12 pure colours of the painter's wheel (yellow at the top, then clockwise yellow-green, green, sky blue, blue, violet, magenta, crimson, red, red-orange, orange, amber) on 5 rings, from the pure colour at the rim to pale tints next to the centre. Every wedge can be picked, red and yellow included; a theme stored in red or yellow is kept. Beside the wheel: the picked colour, a hex field (type a code, then Enter), a preview, the contrast checks of that colour, and, when the flyPad must adjust the colour for the chosen base, the shade it will really use. Nothing changes before **Apply**; **Cancel** keeps the theme as it was. The secondary wheel also offers "Same"; only the hue and colourfulness of a secondary colour count (the base sets its lightness).
- The theme applies to both tablets and to the Payload balance chart. An existing user keeps their theme after the update (the old setting is carried over once).

Light theme fix: several titles and labels were hard-coded white and disappeared on the Light theme (and on a Light base): the Settings page title, the ATC frequency cards and callsign, the checklist and failure (ATA) titles, the preset names, the aircraft symbol of the pushback map, the battery level and the slider labels. They now use the theme text colour; a selected ATC frequency card uses its own text colour.

Limitations and why

Limitation	Why	What to do
Orange, amber or red theme colours look like alerts.	These hues mean caution and warning on the flyPad; the warn and error colours are never themed. The flyPad allows them with a note.	Prefer another hue.
A picked colour can look lighter or darker than its disc or hex code.	Its lightness is moved until the contrast reaches AA on the chosen base.	-

How to use

1. Open **flyPad > Settings > flyPad** and scroll to **Theme**.
2. Press Blue, Dark or Light for a standard theme, or **Custom**.
3. Custom: pick the **Base** (Gray, Black, Light), then a **Primary color** and a **Secondary color** disc, or press the pencil button to pick it on the colour wheel (or type a hex code, then Enter), then **Apply**.
4. Check the contrast list beside the preview card; press **Reset to Blue** to go back to the usual cyan flyPad.

How to test

#	Do this	Expected result
1	Either aircraft, cold and dark at the gate, flyPad on. Note the theme used before the update. Open flyPad > Settings > flyPad, scroll to the bottom.	Theme shows the same theme as before (Blue, Dark or Light), the same colours; no Base / Primary color / Secondary color rows.
2	Press Custom, Base Black, the violet Primary color disc (4th) and the teal Secondary color disc (6th after "Same").	Theme Custom, Base Black; filled buttons and switches violet; the selected tabs (left rail, segmented buttons) teal; the contrast list shows "AA" on every line.
3	Open flyPad > Ground > Payload.	The MTOW line of the balance chart and the "Boarded" seats are violet.
4	Reload the flight, then open Settings > flyPad.	The same custom theme (Black, violet, teal) is back.
5	Primary color: press the pencil button.	The colour wheel: 12 pure colours at the rim (yellow at the top) and 5 rings down to pale tints; no hatched wedge.
6	Press the red wedge of the rim, then Apply.	The note "Close to the amber and red of the alerts: buttons and switches in this color may look like warnings"; after Apply the primary is red. Reload the flight: still red.
7	Pencil button, type 2E86AB and press Enter, then Apply.	No note; the primary becomes a steel blue.
8	Press Reset to Blue.	Theme Blue, the usual cyan flyPad.
9	Choose the Light theme. Open Settings, ATC, Checklists, Failures (an ATA chapter), Presets and Ground > Pushback; look at the battery level in the status bar and at a slider label.	Every title, callsign, preset name, the pushback aircraft symbol, the battery level and the slider labels are dark and readable on the light background (none white).

14. Performance > Buffet: A32NX buffet onset envelope (coffin corner)

> **Aircraft:** A32NX | **Where:** flyPad > Performance > Buffet | **Status:** Tested in the sim by the user: Live page in cruise and in a turn OK, REC MAX 398 as on the MCDU; What if checked from a user screenshot; Live / What if pill row fix OK. Unit tests: 41 (buffet, buffetLive, buffetUnits, RecMaxFlightLevel).

What it is for

A new Performance tab, on the A32NX only: the buffet onset envelope (the coffin corner) from the A320 FCOM LIM-13 BUFFET ONSET chart, for the aircraft now (**Live**) or for any weight, CG, level, Mach and bank (**What if**). It shows how much margin is left before the low speed and high speed buffet, how far the aircraft can bank, the 1.3 g ceiling and the FMS REC MAX.

- The digitised FCOM chart (Mach, pressure altitude, CG, weight) is reduced to one lift table scaled by the ISA pressure ratio, with the chart CG correction (within 1.5 t of the 18 curves; within about 1 % of the QRH M .78 1.3 / 1.4 g lines).
- REC MAX: the FMS value of the MCDU PROG page (A320 FCOM DSC-22_20-40-30: lowest of the 0.3 g buffet margin, MAX CRZ level flight, 300 ft/min at MAX CLB, green dot to VMO/MMO, the certified FL 398). The MCDU now publishes it in `L:A32NX_FM_REC_MAX_FL` (0 without a gross weight) for the flyPad. With one engine out it is the engine-out REC MAX (see the next feature).
- The A380X has no Buffet tab: the A380 FCOM has no buffet onset chart.

What you see

Part	Content

Live / What if	A segmented pill at the top. Live reads the aircraft every second; What if takes typed values (the **A/C** button copies the live values).
Inputs	Gross weight (tag FMS GW, or W&B without an FMS weight; t or klb as set in the flyPad), CG (% MAC), Altitude and Mach (ADR 1, or SIM when ADR 1 is not valid), Bank (IR 1 or SIM).
Chart	Pressure altitude against Mach: the grey 1.0 g line (coffin corner), the cyan 1.3 g line (0.3 g margin), the 1.4 g line (QRH turbulence margin, off by default: chip 1.4 g above the chart), the red VMO / MMO line, the dashed REC MAX line (Live), the aircraft or the entered point; in a turn of 3 degrees or more a dashed amber "NN° bank" line.
Results	The big tile "Buffet margin" (wings level) or "Margin in the turn" with the Level margin (bank 3 degrees or more), then tiles such as Max bank, 1.3 g ceiling, 1.3 g Mach range, REC MAX (Live), Max operating (Mach, MMO or VMO) and 1.0 g ceiling. Margins amber under 0.3 g, red under 0.2 g.
Banners	Red: buffet onset in the turn ("Buffet onset in a 30° turn: max bank 27°..."), level margin below the FMS 0.2 g limit. Amber: level margin below the 0.3 g of REC MAX. Neutral: no gross weight, Mach outside the chart (M .50 to .82).

Pill row fix: the row of the Live / What if pill was 2 px shorter than the pill, so the top of the pill was cut off. The row now takes the height of the pill.

Limitations and why

Limitation	Why	What to do
A32NX only.	No A380 buffet onset chart in the FCOM (its BUFFET CEILING figure is qualitative).	-
No REC MAX tile and line in What if, and none in Live without a gross weight in the FMS.	REC MAX is the FMS value for the aircraft now.	Complete MCDU INIT A and INIT B (ZFW / ZFWCG, block fuel).
Live values from ADR 1 and IR 1.	Design choice: the real flyPad has no ADIRS link; ADR 1 / IR 1 are the captain side sources.	The tag beside each value says where it comes from.
The envelope is advice only.	It is a flyPad design choice built on the FCOM chart, not an Airbus flyPad function.	The FMS REC MAX and the FCOM stay the reference.

How to use

1. Open **flyPad > Performance**, then the **Buffet** tab (5th tab, top right).
2. In cruise, keep **Live**: read the buffet margin, the max bank and REC MAX.
3. To plan a level or a turn, press **What if**, enter Gross weight, CG, Altitude, Mach and Bank, and read the margin and the banner.
4. Turn on the **1.4 g** chip to see the turbulence margin line.

How to test

#	Do this	Expected result
1	A32NX in cruise at FL350 to FL390, about M .78, wings level, autopilot ON, MCDU INIT A and INIT B done. On the MCDU press PROG and note the REC MAX flight level.	A REC MAX FL on the PROG page.

2	flyPad > Performance > Buffet.	Buffet is the 5th tab; Live selected (the pill not cut at the top); Gross weight tagged FMS GW, Altitude and Mach ADR 1, Bank 0 with IR 1; chart with the grey 1.0 g, cyan 1.3 g and red VMO/MMO lines and the aircraft at your FL and Mach; the REC MAX tile and dashed line show the MCDU value; no red banner.
3	Turn the FCU HDG knob 90 degrees left (about 25 degrees of bank) and watch the page.	While the bank is 3 degrees or more: Bank follows the roll, a dashed amber "NN° bank" line, the big tile reads "Margin in the turn", Max bank and Level margin shown. Wings level again: "Buffet margin", no amber line.
4	Press What if. Enter Gross weight 76 t (167.6 klb), CG 24, Altitude 39000, Mach 0.76, Bank 30.	Red banner "Buffet onset in a 30° turn: max bank 27° at FL390, M .760. Level margin 0.13 g is below the FMS 0.2 g limit."; Margin in the turn -0.02 g in red; 1.3 g Mach range "none"; 1.3 g ceiling FL361; no REC MAX tile or line.
5	Press the 1.4 g chip above the chart.	A light cyan 1.4 g line and its legend line appear.
6	Open flyPad > Performance on the A380X.	No Buffet tab.

15. A32NX MCDU PROG: engine-out REC MAX

> **Aircraft:** A32NX | **Where:** MCDU PROG page, REC MAX; flyPad > Performance > Buffet | **Status:** Tested in the sim by the user: OK (user values). Unit test: RecMaxFlightLevel.spec.

What it is for

With one engine out the PROG page REC MAX still showed the all-engines value (about FL380). The FCOM says it shows the engine-out value then.

> **NOTE**
> A320 FCOM, MCDU PROG page : "If one engine is out, this field displays the recommended maximum engine-out altitude, that is computed based on the long-range cruise speed and assuming that anti-ice is off."

When the FMS is in its engine-out condition, REC MAX is now the one engine inoperative long-range cruise ceiling: a straight line through the two FCOM examples, FL182 at 70 t and FL203 at 62 t (PER-OEI-GEN-10 and -15), limited to FL398. With both engines REC MAX is unchanged. The flyPad Buffet page reads the same value. This branch holds only the REC MAX part of the A32NX (the engine, ignition, gravity feed and IR parts are on `core/failures-a32nx`).

Limitations and why

Limitation	Why	What to do
No ISA deviation or anti-ice effect on the engine-out value.	The ceiling table is in the QRH, not in the FCOM text; the FCOM gives ISA examples only, and the PROG value assumes anti-ice off.	Use the QRH for an exact ceiling.

How to use

1. After an engine failure in cruise, open the MCDU PROG page and read REC MAX.

How to test

#	Do this	Expected result

1	A32NX in cruise, both engines running, MCDU INIT A and INIT B done. MCDU PROG page: note REC MAX.	The all-engines value (about FL380).
2	Pedestal: ENG MASTER 2 OFF. MCDU PROG page.	REC MAX about FL180 to FL205 (depends on the weight: about FL182 at 70 t, FL203 at 62 t).
3	flyPad > Performance > Buffet, Live.	The REC MAX tile shows the same engine-out value.
4	ENG MASTER 2 ON (relight), wait for idle. MCDU PROG page.	REC MAX back to the all-engines value.

16. GSX services link: passenger counts, request chips, greyed switch

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Services, GSX services card; Ground > Payload | **Status:** GSX boarding passenger number and counter: tested in the sim by the user: OK (A32NX: GSX given 170 passengers at switch-on, count 0 to 170, Done). Greyed switch, request chips and the deboarding number: not sim-tested. Unit tests: GsxRemote, GsxServicesPanel, ServicesLayout, gsxPassengers, gsxPassengerSync, Material.

What it is for

Fixes of the GSX link of the Services page (the GSX services card, from `core/efb-ground`), found in the sim (A380X at RJTT, GSX Remote API). The link needs the third-party add-on GSX Pro (FSDreamTeam) with its Remote control server on.

- **Passenger number.** GSX boarded its own number of passengers and the Board counter stayed at 0. The flyPad now gives GSX its passenger number (`L:FSDT_GSX_NUMPASSENGERS`, which GSX uses in place of its own figure when set before the boarding or deboarding, GSX manual DEVELOPERS): the passengers planned on the Payload page, or the passengers on board once a deboarding is requested or after landing. The Services page and the Payload pages of both aircraft use the same rule; before, a GSX deboarding started from the Payload page was given 0 passengers.
- **Passenger counter.** The flyPad counts the GSX boarding and deboarding passengers itself: the Board line shows boarded over planned (e.g. 45/120) and its bar fills; before, it showed GSX's own figure twice.
- **Request chips.** A turnaround step offers **Request** only while GSX lets the service be triggered, and **Stop** while it runs. A completed service no longer offers Request (also the Ground equipment rows, e.g. the Catering Truck).
- **Greyed switch.** Without GSX the GSX services switch is faded and does not move. The disabled look of the Material controls is now set by the flyPad itself (the sim browser does not reliably support the CSS :disabled state).

Limitations and why

Limitation	Why	What to do
Needs GSX Pro.	A third-party add-on (FSDreamTeam); the flyPad only drives it.	Install GSX Pro and turn on its Remote control server (GSX settings).
GSX resets its passenger number to 0 when it restarts.	GSX behaviour (GSX manual).	The flyPad gives the number again while the Services page is linked, and just before a Board / Deboard request.

How to use

1. Start GSX Pro with its Remote control server on.
2. Plan the passengers on **flyPad > Ground > Payload** (do not press Board).
3. On **Ground > Services**, turn the **GSX services** switch on when the card says "GSX found".
4. Request the jet bridge or stairs, then **Board > Request** in the Turnaround card; after landing, **Deboard > Request**.

How to test

#	Do this	Expected result
1	Needs GSX Pro (third-party) with its Remote control server on. A380X (or A32NX) at a gate with a jet bridge, GSX not started yet (start the flight before Couatl, or Couatl tray icon > Exit). Open flyPad > Ground > Services.	In the GSX services card: the switch faded and it does not move when tapped; red text "GSX not found: start GSX, with its Remote control server on (GSX settings)".
2	Start GSX (Couatl) and wait.	The card says "GSX found"; the switch is at full brightness and can be turned on.
3	Ground > Payload: set 120 passengers (do not press Board). Back on Services, turn the GSX services switch on.	Within 2 s GSX is given 120 passengers (`L:FSDT_GSX_NUMPASSENGERS` = 120).
4	Ground equipment > Jet Bridge > Request; when connected, Turnaround > Board > Request. Watch the Board line for 2 minutes.	GSX boards 120 passengers; the Board line reads boarded/120, starting near 0 and rising, its bar filling; at the end "Done 120/120"; Ground > Payload shows the same passengers.
5	Look at the Board line, and at Catering (Turnaround and Ground equipment) if catering is done.	No Request chip on a Done service.
6	After the next landing at a gate with GSX: Turnaround > Deboard > Request (Payload page open or not).	GSX is given the passengers on board (not 0) and deboards them; the Deboard line counts them.

17. GSX turnaround: completed services remembered, Baggage steps

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Services, GSX services card > Turnaround | **Status:** Tested in the sim by the user (A380X): Done kept after GSX reset the boarding to available (recording), Baggage row Done (recording: boarding cargo 50 to 100 %), Turnaround horizontal scrollbar gone. Unit tests: gsxTurnaround, GsxServicesPanel.

What it is for

Needs the third-party add-on GSX Pro. GSX gives one completion signal, the service state 6 "completed", and does not keep it: in the sim (A32NX) GSX set the boarding state back to 1 (available) 14 s after the end of the boarding, so the turnaround offered the boarding again as if it had not been done.

- **Completed services remembered.** The flyPad itself remembers the completed boarding, deboarding, catering and refuelling of the current turnaround (also across flyPad page changes). A service requested or running again is forgotten; the boarding is forgotten when a deboarding starts, and the reverse; everything is forgotten at the end of the turnaround (an engine running or the pushback requested). The jet bridge, stairs and GPU follow the sim.
- **Baggage steps.** A Baggage row after Board in the departure turnaround (catering, boarding, baggage, pushback) and after Deboard in the arrival turnaround (deboarding, baggage, catering). GSX has no baggage service to request: it loads the baggage during the boarding and unloads it during the deboarding, so the row has no Request button ("Loaded with the boarding" / "Unloaded with the deboarding" while idle) and shows the GSX baggage progress (`L:FSDT_GSX_BOARDING_CARGO_PERCENT` / `L:FSDT_GSX_DEBOARDING_CARGO_PERCENT`). It is Done when 100 % is reached after the loading was seen running in this turnaround (a 100 % left from an earlier loading does not count), remembered and forgotten with its boarding or deboarding.
- **List fixes.** The Turnaround list scrolls vertically only (no horizontal scrollbar), and the progress bar of a step sits in its own full-width box under the texts (it was pushed out of the card).

Limitations and why

Limitation	Why	What to do

Needs GSX Pro.	A third-party add-on; the states come from its variables and Remote API.	Install GSX Pro with its Remote control server on.
The Done marks last for the current turnaround only.	GSX does not keep its completion; the flyPad clears them when an engine runs or the pushback is requested.	-
No Request on the Baggage rows.	GSX has no separate baggage service.	Request Board or Deboard.

How to use

1. On **Ground > Services**, link GSX with the GSX services switch.
2. Run the turnaround steps in order with their Request chips; a Done step keeps its check mark.
3. Follow the baggage on the Baggage row during the boarding or deboarding.

How to test

#	Do this	Expected result
1	Needs GSX Pro (third-party) with its Remote control server on. A380X or A32NX at a gate, GSX linked on Ground > Services, passengers planned. Look at the Turnaround card.	The departure steps in this order: catering, boarding, Baggage, pushback (service names as GSX gives them); the Baggage row reads "Loaded with the boarding", with no Request button. No horizontal scrollbar in the list.
2	Request Board and watch the Board and Baggage rows.	The progress bars stay inside the card, under the texts; the Baggage progress rises with the GSX baggage loading.
3	Wait for the end of the boarding and the baggage loading.	Board and Baggage both Done with a check mark.
4	Wait about 30 s (GSX resets the boarding to available), then open Ground > Payload and come back to Services.	Board and Baggage are still Done; Board offers no Request.
5	Start an engine (or request the pushback).	The turnaround is cleared: the Done marks go.
6	After landing at a gate with GSX, look at the Turnaround card, then Request Deboard.	The arrival steps: deboarding, Baggage ("Unloaded with the deboarding"), catering; at the end the deboarding and Baggage are Done.

18. GSX refuel progress and jet bridge state

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Services, Ground equipment list (rows "Fuel Truck" and "Jet Bridge") | **Status:** Sim-tested OK by the user on the A380X at RJAA with GSX 4 (2026-10-17): refuel 0 % at the hose connection, then the percentage up and the minutes down; jet bridge docked = Connected + Release, undocking = Leaving, undocked = Request. A32NX not sim-tested. Unit tests: GsxRemote, GsxServicesPanel, useGsxServiceLink, refuelTimeLeft (19 cases).

What it is for

Needs the third-party add-on GSX Pro with its Remote control server on, the GSX services switch on. Two rows of the Ground equipment list showed the wrong GSX state:

- **Fuel Truck.** With the flyPad fuel sync on (flyPad > Settings > 3rd Party Options > GSX Fueling Synchronization) the flyPad pumps the fuel and GSX does not know the target: GSX keeps its own target just above what was pumped, so the row read 99-100 % right after the refuel started. While the flyPad refuel runs, the row now shows the progress towards the flyPad fuel target (fuel received since GSX started / fuel to load) and the whole minutes

left, measured from the fuel flow of the last minute, e.g. "Refueling 25% · 12 min". The minutes appear once a few seconds of flow are measured. Without the flyPad refuel, GSX's own progress stays.

- **Jet Bridge.** GSX 4 reports the jetway (and the stairs) as completed after a dock and after an undock alike; only its phase tells which. The sim does not see a GSX jetway, so the row showed "Done": after an undock with no Request (the jetway could not be called again), and while docked with Request. Now a docked jetway or stairs reads "Connected" with a **Release** chip, "Leaving" while it undocks, and an undocked one offers **Request** again (GSX accepts the call).

Limitations and why

Limitation	Why	What to do
Needs GSX Pro.	A third-party add-on; the states come from its Remote API.	Install GSX Pro with its Remote control server on.
No minutes left during the first seconds of the refuel or while no fuel flows.	The time left comes from the measured fuel flow.	Wait a few seconds after the hose connection.
The fuel progress needs GSX to report its fuel in kg or lb.	The fuel figures come from GSX (its unit setting).	Without them the row shows GSX's own progress.

How to use

1. On **Ground > Services**, link GSX with the GSX services switch.
2. Request the jet bridge; **Release** it before the pushback, **Request** it again if needed.
3. Refuel with the flyPad (fuel target and start on **Ground > Fuel**, GSX fuel sync on) and follow the progress and the minutes left on the Fuel Truck row.

How to test

#	Do this	Expected result
1	Needs GSX Pro (third-party) with its Remote control server on. A380X (or A32NX) at a gate with a jet bridge, GSX linked on Ground > Services, flyPad > Settings > 3rd Party Options > GSX Fueling Synchronization on. Jet Bridge > Request.	Jet Bridge row "Connected" with a Release chip once docked.
2	Ground > Fuel: set a fuel target well above the current fuel and start the refuel. Back on Services, watch the Fuel Truck row.	0 % at the hose connection, then the percentage rises and, after a few seconds, the minutes left count down.
3	Jet Bridge > Release.	"Leaving" while it undocks, then the row offers Request again.
4	Jet Bridge > Request.	The jetway docks again; "Connected" with Release.

19. Taxi page: A380 taxiway width check

> **Aircraft:** A380X | **Where:** flyPad > Ground > Taxi (A380X) | **Status:** Not sim-tested (added; unit tests: taxiWidth.spec.ts; checked in the browser harness on the synthetic test airport XTST with an 18 m taxiway N: the suggestion avoids N, a clearance via N is drawn and flagged).

What it is for

The taxi route page did not know whether a taxiway suits the A380 (wingspan 79.75 m, aerodrome reference code letter F). ICAO Annex 14 Volume I, 3.9.3 (taxiway width of the straight portion) gives 25 m for code letter F. On the

A380 flyPad, each taxi line of the Navigraph airport map is now rated:

- **Wingspan limit of the line** in the airport map, when the map has one, against 79.75 m (not seen in the Navigraph data so far; read only when present).
- **Else the pavement width under the line:** measured across the line (60 m each side) through the taxiway, apron and runway areas of the map (adjacent areas joined, holes left out), the narrowest of samples every 30 m away from the junction fillets, against 25 m with 1 m tolerance.
- **Else "not verified"** (no data), which is not treated as unsafe.

Situation	What you should see
SUGGEST, or the proposed runway entry	The route keeps off the lines too narrow when it can.
A typed clearance that needs a line too narrow	The route still follows the clearance and is drawn, but flagged: the route text marks the line, and a chip and a banner list the narrow taxiways (ATC may know better: the crew decides).
The map	Lines too narrow in orange, lines not verified dashed grey, with a legend.
A32NX	No width check (the A320 is code C).

Limitations and why

Limitation	Why	What to do
Pavement width is not wingtip clearance.	Buildings and parked aircraft are not in the check.	Follow the airport charts.
Lines on aprons measure wide.	The apron area around them counts as pavement.	-
An airport may clear the A380 on narrower taxiways with restrictions.	The flag means "check the airport A380 notes", not "forbidden".	Check the airport A380 notes.
Airport loading may take longer.	The runway intersection layer is now loaded for the pavement, and every line is measured.	-

How to use

1. A380X flyPad > Ground > Taxi: plan the route as usual.
2. Look at the map legend: orange lines are too narrow, dashed grey ones not verified.
3. If the route chip or banner lists narrow taxiways, check the airport A380 notes or ask ATC for another route.

How to test

#	Do this	Expected result
1	A380X at RJTT (Navigraph linked). flyPad > Ground > Taxi, Departure, airport RJTT.	The map shows a legend; some lines in orange (too narrow) or dashed grey (not verified).
2	SUGGEST a route to a runway.	The route avoids the orange lines when another way exists.
3	Type a clearance through an orange taxiway.	The route follows it and is drawn; the route text marks the line; a chip and a banner list it.
4	Same page on the A32NX.	No width colours, no banner.

What PR testers should check

- At RJTT (A380): which taxiways turn orange or dashed, and whether known A380 routes show as too narrow.

- The airport load time.
- Not sim-tested yet.

20. A380X smart FIVE HUNDRED pin option back in the flyPad

> **Aircraft:** A380X | **Where:** flyPad > Settings > Aircraft Options / Pin Programs > Automatic Call Outs | **Status:** Not sim-tested (added; the callout logic it sets has a unit test on `core/a380x-elec-fuel-alerts`).

What it is for

The A380 FCOM automatic callouts (DSC-34-10-50) have a "smart FIVE HUNDRED" pin programming: the FIVE HUNDRED callout is given only when the aircraft is more than half a dot off the glide slope, or without a valid glide slope. The flyPad option was commented out. The Automatic Call Outs page shows it again, as a sub-option **Inhibited When Glideslope Captured** under **Five Hundred** (greyed while Five Hundred is off).

Limitations and why

Limitation	Why	What to do
The option does nothing without `core/a380x-elec-fuel-alerts`.	The FWS callout that reads the pin is on that branch.	Merge both.

How to use

1. flyPad > Settings > Aircraft Options / Pin Programs > Automatic Call Outs: Five Hundred on, then **Inhibited When Glideslope Captured** on.

How to test

#	Do this	Expected result
1	A380X flyPad > Settings > Aircraft Options / Pin Programs > Automatic Call Outs.	Under Five Hundred the sub-option Inhibited When Glideslope Captured; greyed while Five Hundred is off.
2	Turn both on; fly an ILS approach on the glide slope (with `core/a380x-elec-fuel-alerts`).	No FIVE HUNDRED callout at 500 ft.
3	Repeat more than half a dot off the glide slope (or a non-ILS approach).	FIVE HUNDRED is called.

What PR testers should check

- The option is saved and greyed correctly; the callout follows it with the FWS branch merged.
- Not sim-tested yet.